

CHAPTER 4 Designing Studies

4.2 Experiments

The Practice of Statistics, 5th Edition
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Experiments

Learning Objectives

After this section, you should be able to:

- ✓ DISTINGUISH between an observational study and an experiment.
- ✓ EXPLAIN the concept of confounding.
- ✓ IDENTIFY the experimental units, explanatory and response variables, and treatments in an experiment.
- ✓ EXPLAIN the purpose of comparison, random assignment, control, and replication in an experiment.
- ✓ DESCRIBE a completely randomized design for an experiment.
- ✓ DESCRIBE the placebo effect and the purpose of blinding in an experiment.
- ✓ INTERPRET the meaning of statistically significant in the context of an experiment.
- ✓ EXPLAIN the purpose of blocking in an experiment. DESCRIBE a randomized block design or a matched pairs design for an experiment.

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2

Observational Study vs. Experiment

An **observational study** 观测研究 observes individuals and measures variables of interest but does not attempt to influence the responses. 观测数据但不干扰，目的是描述某个群体或某种情况，被动的收集数据。
- sample survey

An **experiment** 实验 deliberately imposes some treatment on individuals to measure their responses.
在实验中，不同的条件（treatment）会被分配给实验单位（参与者，受试者）。目的是研究改特定的处理是否会使其发生变化，是主动产生数据。

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3

The language of Experiments

An experiment is a statistical study in which we actually do something (a **treatment** 处理后给予的条件) to people, animals, or objects (the **experimental units** 实验单元 or **subjects** 实验对象) to observe the response.

DEFINITION Treatment, Experimental unit, Subjects

A specific condition applied to the individuals in an experiment is called a **treatment**. If an experiment has several explanatory variables, a treatment is a combination of specific values of these variables. An **experimental unit** is the object to which a treatment is randomly assigned. When the experimental units are human beings, they are often called **subjects**.

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4

Treatment, Experimental unit, Subjects

Malaria 疟疾 causes hundreds of thousands of deaths each year, with many of the victims being children. Will regularly screening children for the malaria parasite and treating those who test positive reduce the proportion of children who develop the disease?

Researchers worked with children in 101 schools in Kenya, randomly assigning half of the schools to receive regular screenings and follow-up treatments and the remaining schools to receive no regular screening. Children at all 101 schools were tested for malaria at the end of the study.

Identify the **treatments** and the **experimental units** in this experiment.

Explanatory variable: _____

From Chapter 3: A **response variable** 响应变量 measures an outcome of a study. 在给定 treatment 之后的结果。An **explanatory variable** 解释变量 may help explain or predict changes in a response variable. 自变量

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Multiple explanatory variables: Factor 因子 (等于自变量), Levels (factor 的等级)

Have you ever dropped a tasty piece of food on the ground, then quickly picked it up and eaten it? If so, you probably thought about the "five-second rule," which states that a piece of food is safe to eat if it has been on the floor less than 5 seconds. The rule is based on the belief that bacteria need time to transfer from the floor to the food. But does it work?

Researchers from Rutgers University put the five-second rule to the test. They used **four different types of food**: watermelon, bread, bread with butter, and gummy candy. They dropped the food onto **four different surfaces**: stainless steel 不锈钢, ceramic tile 瓷砖, wood 木头, and carpet 地毯. And they waited for **four different lengths of time**: less than 1 second, 5 seconds, 30 seconds, and 300 seconds. Finally, they used bacteria prepared **two different ways**: in a tryptic soy broth 高汤 or peptone buffer 缓冲液. Once the bacteria were ready, the researchers spread them out on the different surfaces and started dropping food.

If the researchers used every possible combination to form the treatments, how many treatments were included in the experiment?

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6

Factor, Levels

DEFINITION Factor, Levels

In an experiment, a **factor** is a variable that is manipulated and may cause a change in the response variable. The different values of a factor are called **levels**.

Experiment: Investigating the effects of different water conditions on the growth rate of plants.

1. **Factor:** Watering amount
2. **Levels:** High, medium, low

Experiment: Investigating the effects of different water and sunlight conditions on the growth rate of plants.

1. **Factors:** Watering amount and sunlight exposure
2. **Levels for Watering:** High, medium, low
3. **Levels for Sunlight:** Abundant, medium, scarce

Does caffeine affect pulse rate?

Many students regularly consume caffeine to help them stay alert. So it seems plausible that **taking caffeine might increase an individual's pulse rate**. Is this true? One way to investigate this claim is to ask volunteers to measure their pulse rates, drink some cola with caffeine, measure their pulse rates again after 10 minutes, and calculate the increase in pulse rate.

Here is the design:

Experimental units → Treatment → Measure response

Students → Cola with caffeine → Change in pulse rate

Result: The pulse rate of every student went up

Confounding 干扰变量

Well-designed experiments take steps to prevent confounding.

Confounding 干扰变量/混淆变量/潜在变量 occurs when two variables are associated in such a way that their effects on a response variable cannot be distinguished from each other.

干扰变量是与 explanatory variable (x) 相关的量, 会影响 response variable (y)。

Experimental units → Treatment → Measure response
Students → Cola with caffeine → Change in pulse rate

最小二乘回归, explanatory variable 和 response variable 的相关性 correlation, 而非因果关系 causation, 因为 confounding variable 的存在可能会影响结果。

Comparative Experiment

Experimental units → Treatment → Measure response
Students → Cola with caffeine → Change in pulse rate

Revised Experiment: Does caffeine affect pulse rate?

• **Objective:** Determine if caffeine increases an individual's pulse rate.

• **Procedure:**

1. **Selection:** Begin by selecting a group of students. Once selected, assign these students into two groups: Control Group and Treatment Group.

2. **Control Group 对照组/控制组:** This group will drink a cola without caffeine (**placebo 安慰剂**). 最贴近现实的组, 不改变任何因素的组

3. **Experimental Group:** This group will drink a cola with caffeine.

Designing Experiments: Control Group 控制组

DEFINITION Control group

In an experiment, a **control group** is used to provide a baseline for comparing the effects of other treatments. Depending on the purpose of the experiment, a control group may be given an inactive treatment (placebo), an active treatment, or no treatment at all.

- 控制组/对照组的目的是提供一个基线, 以便于比较实验组的效果;
- 有助于消除干扰因素对实验结果的影响;
- 最贴近现实的组。

Designing Experiments: Placebo 安慰剂

Placebo

A placebo is a treatment that has no active ingredient, but is otherwise like other treatments.



DEFINITION Placebo effect

The **placebo effect** describes the fact that some subjects in an experiment will respond favorably to any treatment, even an inactive treatment.

病人虽然获得无效的治疗, 但却“预料”或“相信” expectation of getting better 治疗有效, 而让病患症状得到舒缓的现象。

<https://www.youtube.com/watch?v=ih16hSI09Fo>

Designing Experiments: Placebo effect 安慰剂效应

Revised Experiment: Does caffeine affect pulse rate?

•**Objective:** Determine if caffeine increases an individual's pulse rate.

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Designing Experiments: Placebo effect 安慰剂效应

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3.**Experimental Group**: This group will drink a cola with caffeine.

4.**Blinding**: Neither the students nor the individuals measuring the pulse rates should know which group received the cola with caffeine and which received the placebo. This is called a **double-blind experiment 双盲实验**.

5.**Baseline Measurement**:

6.**Post-treatment Measurement**: After drinking the cola, measure the pulse rates of all participants again after 10 minutes.

7.**Analysis**: Calculate the difference in pulse rate before and after for each participant and compare the average change between the two groups.

●● Double-blind 双盲实验, Single-blind 单盲实验

DEFINITION Double-blind, Single-blind

In a **double-blind** experiment, neither the subjects nor those who interact with them and measure the response variable know which treatment a subject received.

In a **single-blind** experiment, either the subjects don't know which treatment they are receiving or the people who interact with them and measure the response variable don't know which subjects are receiving which treatment.

- 双盲实验: 实验对象、和实验对象有互动的人、研究团队都不知道 treatment 是如何分配的 (只有极少数设计实验的人知道)
- 单盲实验: 实验对象不知道他们接受的 treatment 是什么, 但是研究团队知道。

Smoking and ADHD

Problem: In a study of more than 4700 children, researchers from Cincinnati Children's Hospital Medical Center found that those children whose mothers smoked during pregnancy were more than twice as likely to develop ADHD as children whose mothers had not smoked. [Explain how confounding makes it unreasonable to conclude that a mother's smoking during pregnancy causes an increase in the risk of ADHD in her children based on this study.](#)

Solution: It is possible that the mothers who smoked during pregnancy were also more likely to have unhealthy diets. If people with unhealthy diets are also more likely to have children with ADHD, then it could be that unhealthy diets caused the increase in ADHD risk, not smoking.

How to Experiment Well

Revised Experiment: Does caffeine affect pulse rate?

•**Objective:** Determine if caffeine increases an individual's pulse rate.

•**Procedure:**

1.**Selection** : Begin by selecting a group of students. Once selected, assign these students into two groups: Control Group and Treatment Group.

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How to Experiment Well

Revised Experiment: Does caffeine affect pulse rate?

•**Objective:** Determine if caffeine increases an individual's pulse rate.

•**Procedure:**

1.**Random Selection/Assignment** : Begin by randomly selecting a group of students. Once selected, randomly assign these students into two groups: Control Group and Treatment Group.

2.**Control Group 对照组/控制组**: This group will drink a cola without caffeine (**placebo 安慰剂**). 最贴近现实的组, 不改变任何因素的组

3.**Experimental Group**: This group will drink a cola with caffeine.

4.**Blinding**: Neither the students nor the individuals measuring the pulse rates should know which group received the cola with caffeine and which received the placebo. This is called a double-blind experiment **双盲实验**.

5.**Baseline Measurement**:

6.**Post-treatment Measurement**: After drinking the cola, measure the pulse rates of all participants again after 10 minutes.

7.**Analysis**: Calculate the difference in pulse rate before and after for each participant and compare the average change between the two groups.

Designing Experiments: Random Assignment

To create roughly equivalent groups at the beginning of an experiment, we use **random assignment** 随机分配 to determine which experimental units get which treatment.

DEFINITION Random assignment

In an experiment, **random assignment** means that experimental units are assigned to treatments using a chance process.

A total of 20 students have agreed to participate in an experiment comparing the effects of caffeinated cola and caffeine-free cola on pulse rates. Describe how you would randomly assign 10 students to each of the two treatments:

- Using 20 identical slips of paper
- Using technology
- Using Table D

Designing Experiments: Random Assignment

A total of 20 students have agreed to participate in an experiment comparing the effects of caffeinated cola and caffeine-free cola on pulse rates. Describe how you would randomly assign 10 students to each of the two treatments:

- Using 20 identical slips of paper
- Using technology
- Using Table D

(a)

When describing a method of random assignment, don't stop after creating the groups. Make sure to identify which group gets which treatment.

Designing Experiments: Random Assignment

A total of 20 students have agreed to participate in an experiment comparing the effects of caffeinated cola and caffeine-free cola on pulse rates. Describe how you would randomly assign 10 students to each of the two treatments:

- Using 20 identical slips of paper
- Using technology
- Using Table D

(b)

When using a random number generator or a table of random digits to assign treatments, make sure to account for the possibility of repeated numbers when describing your method.

Designing Experiments: Random Assignment

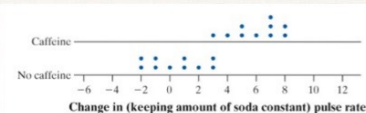
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- Using Table D

(c)

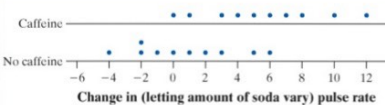
111	81486	69487	60513	09297	00412	71238	27649	39950
112	59636	88804	04634	71197	19352	73089	84898	45785
113	62568	70206	40325	03699	71080	22553	11486	11776
114	45149	32992	75730	66280	03819	56202	02938	70915
115	61041	77684	94322	24709	73698	14526	31893	32592

How to Experiment Well



The results of an experiment in which the amount of cola consumed was the same for all participating students.

We also want to **control** other variables to reduce the variability in the response variable.



The results of an experiment in which the students were permitted to choose how much or how little cola they consumed.

How to Experiment Well

- Would you trust an experiment with just one student in each group?

DEFINITION Replication

In an experiment, **replication** means using enough experimental units to distinguish a difference in the effects of the treatments from chance variation due to the random assignment.

- 指某一特定处理可以对一个实验单位重复进行很多次。
- 由于实验的个体差异以及其影响因素的存在，同一处理对不同实验单元所产生的效果也有差异。通过一定数量的重复实验，该处理的真实效应就会比较确定地显现出来。

▲ In statistics, replication means "use enough subjects." In other fields, the term replication has a different meaning.

Principles of Experimental Design

Principles of Experimental Design

The basic principles for designing experiments are as follows:

1. **Comparison.** Use a design that compares two or more treatments.
2. **Random assignment.** Use chance to assign experimental units to treatments. Doing so helps create roughly equivalent groups of experimental units by balancing the effects of other variables among the treatment groups.
3. **Control.** Keep other variables that might affect the response the same for all groups.
4. **Replication.** Use enough experimental units in each group so that any differences in the effects of the treatments can be distinguished from chance differences between the groups.

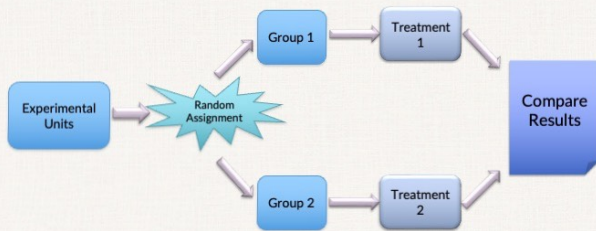
Randomization 随机化

- Type1: If each **experimental unit** is supposed to receive **only one treatment**, then which experimental unit receives which treatment should be determined randomly.
 - A total of 20 students have agreed to participate in an experiment comparing the effects of caffeinated cola and caffeine-free cola on pulse rates.
- Type2: If each **experimental unit** is supposed to receive **all treatments**, then the order of the treatment should be determined randomly for each experimental unit.
 - Consider an experiment where a single individual is to test three different types of headphones (A, B, and C) to rate their comfort.

Completely Randomized Design 完全随机化设计

In a **completely randomized design**, the treatments are assigned to all the experimental units **completely by chance**.

Some experiments may include a control group that receives an inactive treatment or an existing baseline treatment.

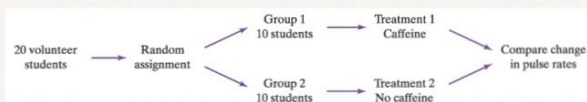


Completely Randomized Design 完全随机化设计

- A total of 20 students have agreed to participate in an experiment comparing the effects of caffeinated cola and caffeine-free cola on pulse rates.
- Describe the design of possible experiment. You are expected to describe how treatments are assigned to the experimental units and clearly state what will be measured or compared.

AP[®] EXAM TIP If you are asked to describe the design of an experiment on the AP[®] exam, you won't get full credit for providing only a diagram like Figure 4.5. You are expected to describe how the treatments are assigned to the experimental units and to clearly state what will be measured or compared. Some students prefer to start with a diagram and then add a few sentences. Others choose to skip the diagram and put their entire response in narrative form.

Flipping Coins



Suppose we ask each student in the caffeine experiment to toss a coin. If it's heads, then the student will drink the cola with caffeine. If it's tails, then the student will drink the caffeine-free cola.

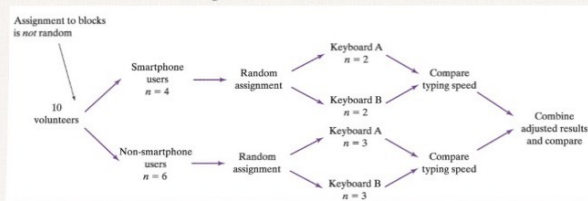
AP[®] EXAM TIP

If you are asked to describe a completely randomized design, stay away from flipping coins.

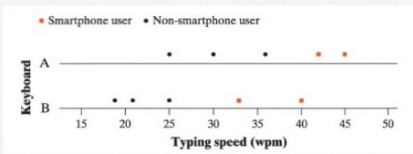
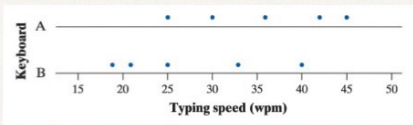
Randomized Block Designs 随机区域设计

Suppose that a mobile phone company is considering two different keyboard designs (A and B) for its new smartphone. The company decides to perform an experiment to compare the two keyboards using a group of 10 volunteers. The response variable is typing speed, measured in words per minute.

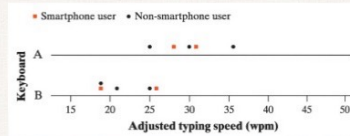
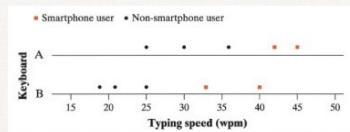
Completely Randomized Design Randomized Block Design



Each of the 10 volunteers typed the same passage and the typing speed was recorded. Here are the results:



In fact, the average typing speed for the **smartphone users** was 40, while the average typing speed for **non-smartphone users** was only 26, a difference of 14 words per minute.



Randomized Block Designs 随机区域设计

DEFINITION Block, Randomized block design

A **block** is a group of experimental units that are known before the experiment to be similar in some way that is expected to affect the response to the treatments.

In a **randomized block design**, the random assignment of experimental units to treatments is carried out separately within each block.

Control what you can, **block** on what you can't control, and randomize to create comparable groups.

- A block is a group of **homogeneous** experimental units 区组是一组具有相同性质的实验单位。他们在实验前就被认定在某种程度上会影响到实验结果。
- 区组设计通过把外变量（混淆变量）引入实验中构成区组，以控制这些外在变量（混淆变量）的影响。

Matched Pairs Design 配对设计

MATCHED PAIRS DESIGN :

FIRST STEP :

Match each participant with its closest partner



SECOND STEP :

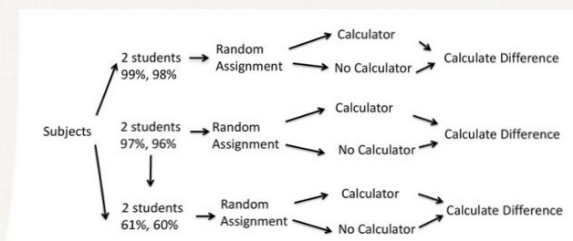
1 participant from each pair gets randomly assigned, and the other is automatically assigned to the other group



- A common type of randomized block design for comparing two treatments is a matched pairs design.
- The idea is to create blocks by matching **pairs of similar** experimental units.
- 本质上是一种特殊的Block Design
- 通过将类似的实验单位（例如，受试者）配对，以减少实验误差。特别是当受试者之间的差异可能对结果产生显著影响时。

Matched Pairs Design 配对设计

Imagine an experiment designed to test the effect of using a calculator on an Algebra 1 test. Each subject is paired with another student who has a very similar grade in the class prior to taking the test.



Language of ...

Experiments:

- Treatment
- Experimental Unit
- Subject
- Factor
- Level
- Control
- Control Group
- Placebo
- Blinding
- Replication
- Random Assignment/ Completely Randomized Design
- Randomized Block Design / Block
- Matches Pairs Design

How to Choose an SRS

Choosing an SRS With Technology

Step 1: **Label.** Give each individual in the population a distinct numerical label from 1 to N.

Step 2: **Randomize.** Use a random number generator to obtain n different integers from 1 to N.

TI-83/84

- Press **MATH**, then select PRB and **randInt**(. Complete the command **randInt**(1,1750) and press **ENTER**.

How to Choose an SRS

How to Choose an SRS Using Table D

Step 1: **Label.** Give each member of the population a numerical label with the same number of digits. 对抽样对象中每一个个体指定一个数字编码，保证每个编码是相同位数 Use as few digits as possible.

Step 2: **Randomize.** Read consecutive groups of digits of the appropriate length from left to right across a line in Table D. Ignore any group of digits that wasn't used as a label or that duplicates a label already in the sample. Stop when you have chosen n different labels.

Your sample contains the individuals whose labels you find.

Table D Random digits								
LINE								
101	19223	95034	05756	28713	96409	12531	42544	82853
102	73676	47150	99400	01927	27754	42648	82425	36290
103	45467	71709	77558	00095	32863	29485	82226	90056

Stratified sampling



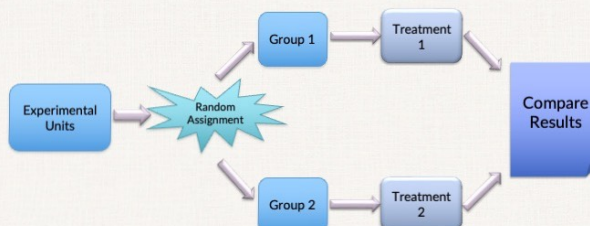
Cluster sampling



Completely Randomized Design 完全随机化设计

In a **completely randomized design**, the treatments are assigned to all the experimental units completely by chance.

Some experiments may include a control group that receives an inactive treatment or an existing baseline treatment.



Language of ...

"Use a randomized block design to select the sample for this survey"

"Use random sampling to select sample/participants for this survey."

"This experiment suffers from nonresponse because some subjects dropped out during the study"

"This experiment has issues with **attrition** (participants leaving a study) because some subjects dropped out during the study."

"For our observational study on the effects of smoking, we had a control group of individuals who were given a placebo cigarette."

"observational study...we compared smokers with non-smokers."

"In our survey on consumer preferences, we blinded participants so they wouldn't know which brand they were assessing."

"survey... participants were asked to evaluate different brands based on their experiences."

"In our experiment to determine the best teaching method, we observed classrooms without intervening or making any changes."

"In our study to determine the best teaching method, we observed..."

Inference for Experiments

In an experiment, researchers usually hope to **see a difference in the responses so large** that it is unlikely to happen just because of chance variation.

We can use the laws of **probability**, which describe chance behavior, to learn whether the treatment effects are larger than we would expect to see if only chance were operating.

If they are, we call them **statistically significant**.

An observed effect so large that it would rarely occur by chance is called **statistically significant**.

A statistically significant association in data from a **well-designed experiment** does imply causation.

●● Principles of Experimental Design

Principles of Experimental Design

The basic principles for designing experiments are as follows:

1. **Comparison.** Use a design that compares two or more treatments.
2. **Random assignment.** Use chance to assign experimental units to treatments. Doing so helps create roughly equivalent groups of experimental units by balancing the effects of other variables among the treatment groups.
3. **Control.** Keep other variables that might affect the response the same for all groups.
4. **Replication.** Use enough experimental units in each group so that any differences in the effects of the treatments can be distinguished from chance differences between the groups.

Experiments

Section Summary

In this section, we learned how to...

- ✓ DISTINGUISH between an **observational study** and an **experiment**.
- ✓ EXPLAIN the concept of **confounding**.
- ✓ IDENTIFY the **experimental units**, explanatory and response variables, and **treatments** in an experiment.
- ✓ EXPLAIN the purpose of **comparison, random assignment, control, and replication** in an experiment.
- ✓ DESCRIBE a **completely randomized design** for an experiment.
- ✓ DESCRIBE the **placebo effect** and the purpose of **blinding** in an experiment.
- ✓ INTERPRET the meaning of **statistically significant** in the context of an experiment.
- ✓ EXPLAIN the purpose of **blocking** in an experiment.
- ✓ DESCRIBE a **randomized block design** or a **matched pairs design** for an experiment.